PROJECT 1

SALES ANALYTICS DASHBOARD

Interactive Sales Intelligence Platform

Chandra Harshitha | B.Tech - Computer Science and Engineering

SALES ANALYTICS DASHBOARD

Internship Project Report

Interactive Sales Intelligence Platform for Revenue, Profitability, Product, Customer and Regional Analysis

Developed By	Chandra Harshitha
Degree	B.Tech – Computer Science and Engineering
Project Domain	Data Analytics / Business Intelligence
Dataset	5,000 sales transaction records
Technology	Python, Pandas, Plotly, Streamlit
GitHub	github.com/Harshi1904/Sales_Analytics_Dashboard
Live Dashboard	harshithasalesanalyticsdashboard.streamlit.app

Prepared for internship submission, portfolio presentation and technical evaluation.

1. Project Overview

The Sales Analytics Dashboard is an interactive business intelligence application that transforms transactional sales data into decision-ready insights. It combines data processing, KPI calculation and interactive visualization in a multipage Streamlit platform.

2. Problem Statement

Raw sales transactions are difficult to interpret when viewed as spreadsheets or CSV records. Business users need a faster way to understand revenue, profitability, product performance, customer contribution, regional performance and growth. This project addresses that need through an interactive analytics dashboard.

3. Project Objectives

- Develop an interactive sales analytics platform using Python and Streamlit.
- Prepare transactional data for reliable business analysis.
- Provide executive KPIs for sales, profit, orders, customers, quantity and margin.
- Analyze monthly sales, profit and month-over-month growth.
- Compare product, category, customer, segment and regional performance.
- Provide filters for date, region, category and customer segment.
- Enable filtered CSV export for further analysis.
- Generate automated business insights from the selected data.

4. Dataset Description

The application uses a synthetic sales transaction dataset containing 5,000 records. It covers order details, customers, geography, products, pricing, costs, profit and operational information.

Area	Representative Fields
Order & Time	Order_ID, Order_Date, Ship_Date, Order_Month, Order_Year

Customer	Customer_ID, Customer_Name, Segment
Geography	Region, State, City
Product	Category, Sub_Category, Product_Name
Transaction	Quantity, Unit_Price, Discount, Sales, Cost, Profit
Operations	Salesperson, Shipping_Days
Derived	Profit_Margin

5. Technology Stack

Technology	Purpose
Python	Core programming and analytics logic
Pandas	Data loading, transformation, grouping and aggregation
Plotly	Interactive charts and visual analytics
Streamlit	Interactive multipage web dashboard
CSV	Transactional data storage and filtered export
Git & GitHub	Version control and project hosting

6. System Workflow & Architecture

The application follows a structured workflow from raw transactions to business insight: load the CSV, prepare the data, apply filters, calculate KPIs, visualize results, interpret the findings and export the filtered data.

Step	Process	Description
1	Load	Read the CSV dataset using Pandas.
2	Prepare	Convert date fields and prepare analytical dimensions.
3	Filter	Apply date, region, category and segment controls.
4	Calculate	Compute KPIs, growth, sales, profit and customer/product metrics.
5	Visualize	Render interactive Plotly charts and tables.
6	Interpret	Generate automated observations from the selected data.
7	Export	Download the filtered dataset as CSV.

Application Modules

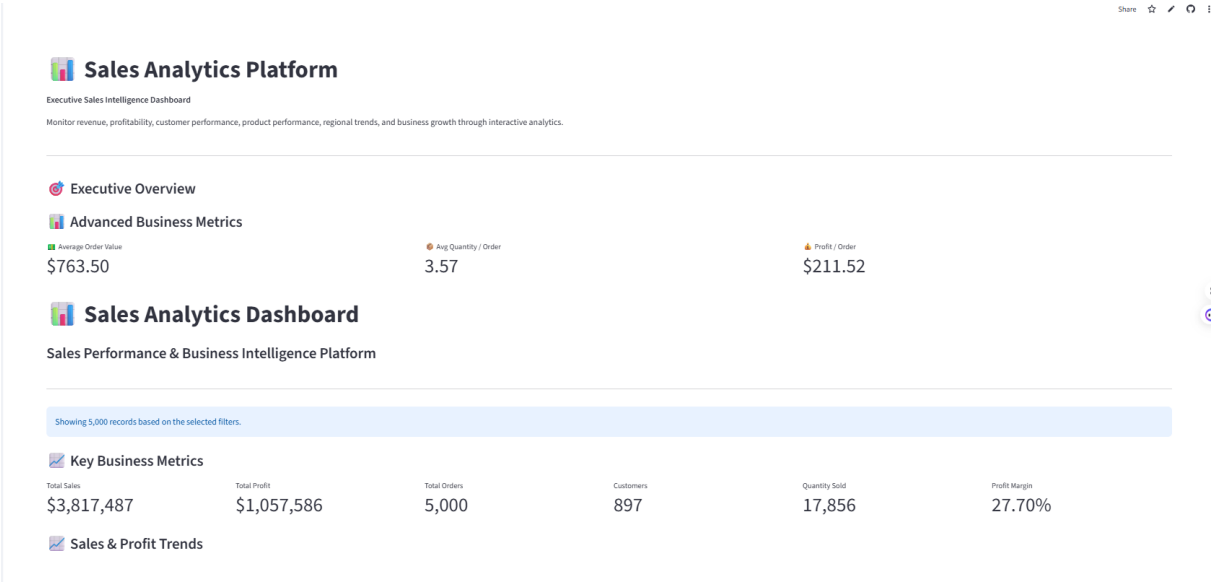
- Landing Page – project introduction and module navigation.
- Executive Overview – KPIs, advanced metrics, monthly performance and growth.
- Business Performance – sales/profit trends, category analysis and regional analysis.
- Product Analytics – top/bottom products, profitability and category/sub-category analysis.
- Customer Analytics – top customers, customer value and segment performance.
- Business Insights – automated observations and filtered-data summary.

7. Key Performance Indicators

- Total Sales, Total Profit, Total Orders, Customers, Quantity Sold and Profit Margin.
- Average Order Value, Average Quantity per Order and Profit per Order.
- Monthly Sales Growth and Monthly Profit Growth for trend monitoring.

8. Dashboard Implementation

8.1 Landing Page & Executive Overview



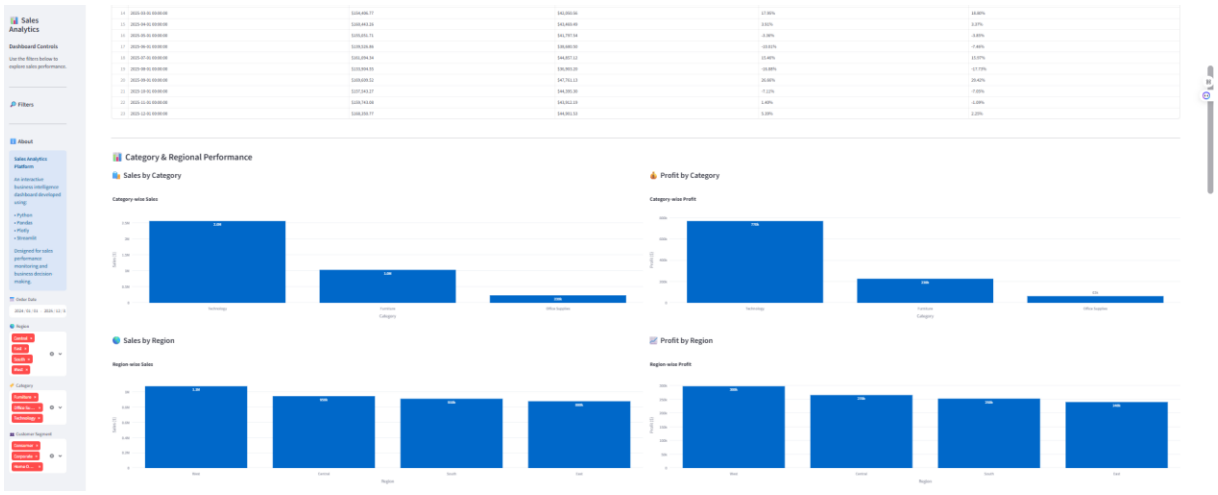
Landing Page & Executive Overview

8.2 Sales & Profit Trends and Growth Analysis



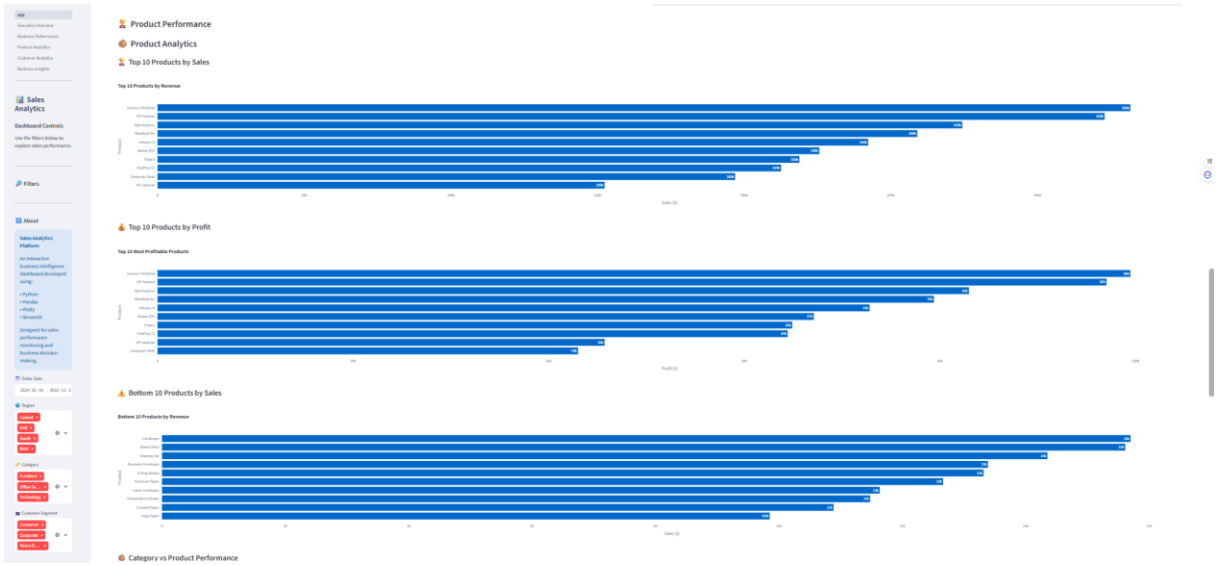
Sales & Profit Trends and Growth Analysis

8.3 Category & Regional Performance



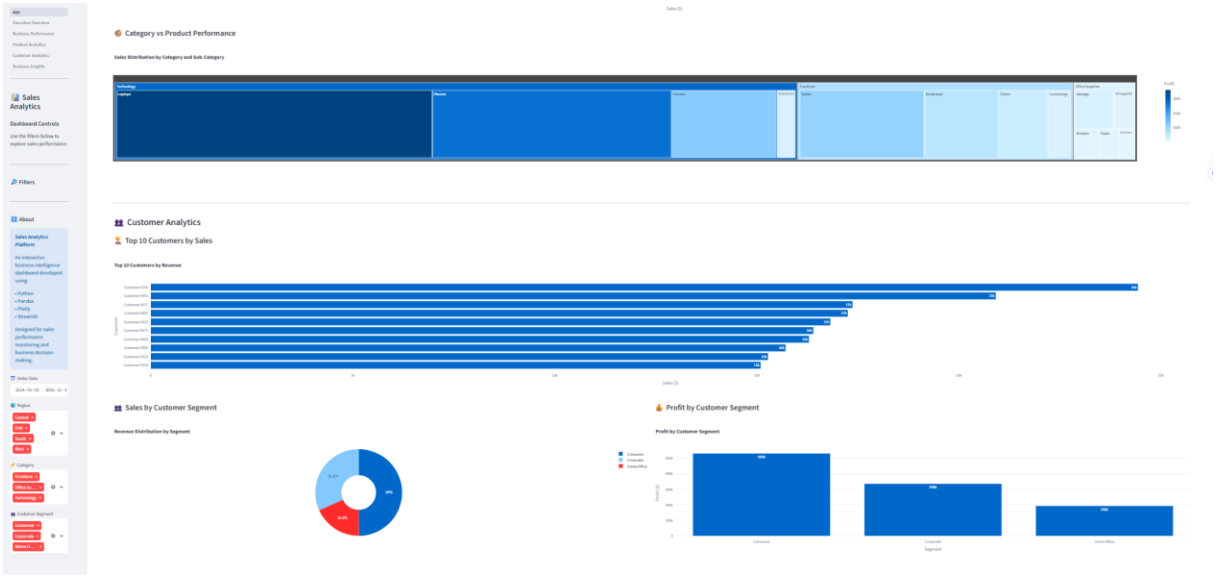
Category & Regional Performance

8.4 Product Analytics



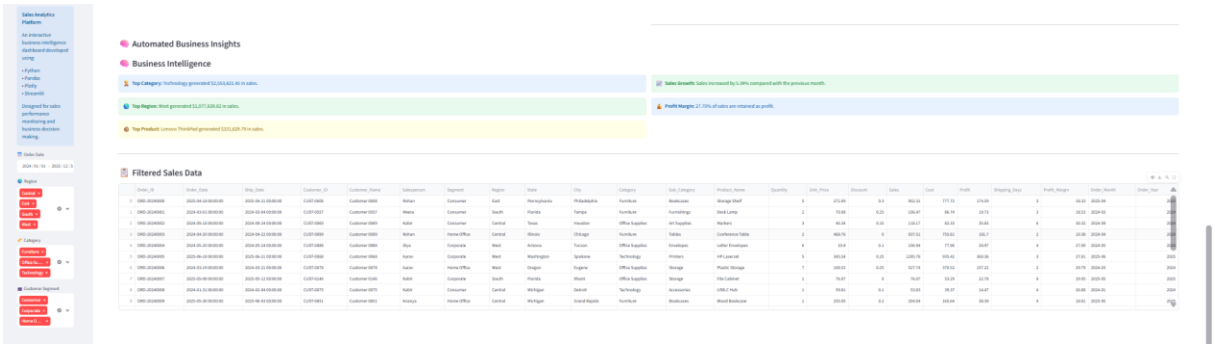
Product Analytics

8.5 Customer Analytics



Customer Analytics

8.6 Business Insights & Filtered Data



Business Insights & Filtered Data

9. Key Business Findings

In the captured full-dataset dashboard view, the platform reports approximately $3.82M in sales, $1.06M in profit, 5,000 orders, 897 customers, 17,856 units sold and a 27.70% profit margin. The captured insights identify Technology as the leading category, West as the leading region, and Lenovo ThinkPad as the top product by sales.

10. Business Value

- Provides fast executive-level monitoring of revenue and profitability.
- Highlights high-performing and low-performing products.
- Supports category and regional comparison.
- Helps identify high-value customers and customer segments.
- Makes monthly growth patterns easier to monitor.
- Enables focused analysis through interactive filters.
- Allows filtered transactions to be exported for further work.

11. Challenges & Solutions

Challenge	Solution
Large transaction table is difficult to interpret.	Converted transactions into KPIs, grouped summaries and interactive charts.
Different users need different views.	Created separate multipage analytical modules.
Users need focused analysis.	Added date, region, category and customer-segment filters.
Users need data outside the dashboard.	Added filtered CSV download.
No reliable latitude/longitude in the dataset.	Used city and regional analysis instead of misleading map coordinates.

12. Future Scope

Future versions can include sales forecasting, anomaly detection, customer lifetime value, cohort analysis, automated email reports, database connectivity, scheduled refresh, role-based access and a true geographic map layer when latitude/longitude data is available.

13. Conclusion

The Sales Analytics Dashboard demonstrates how Python-based analytics can transform transactional sales data into an interactive business intelligence application. The project integrates data preparation, KPI computation, interactive visualization, filtering, export and automated insight generation in a structured multipage Streamlit platform.

14. Project Links

GitHub Repository	github.com/Harshi1904/Sales_Analytics_Dashboard
Live Dashboard	harshithasalesanalyticsdashboard.streamlit.app

Developed By: Chandra Harshitha
B.Tech – Computer Science and Engineering
Data Analytics / Business Intelligence Project

PROJECT 2

HR ANALYTICS DASHBOARD

Intraniz Internship Project

Chandra Harshitha | B.Tech - Computer Science and Engineering

HR Analytics Dashboard

Intraniz Internship Project Report

Presented by
Chandra Harshitha

Workforce Intelligence • Employee Insights • Attrition Analysis

1. Project Overview

Objective: Develop an interactive HR analytics dashboard to understand workforce composition, employee attrition and employee-related patterns.

Dataset: IBM HR Analytics Employee Attrition & Performance dataset — 1,470 employee records and 35 features.

Technology Stack

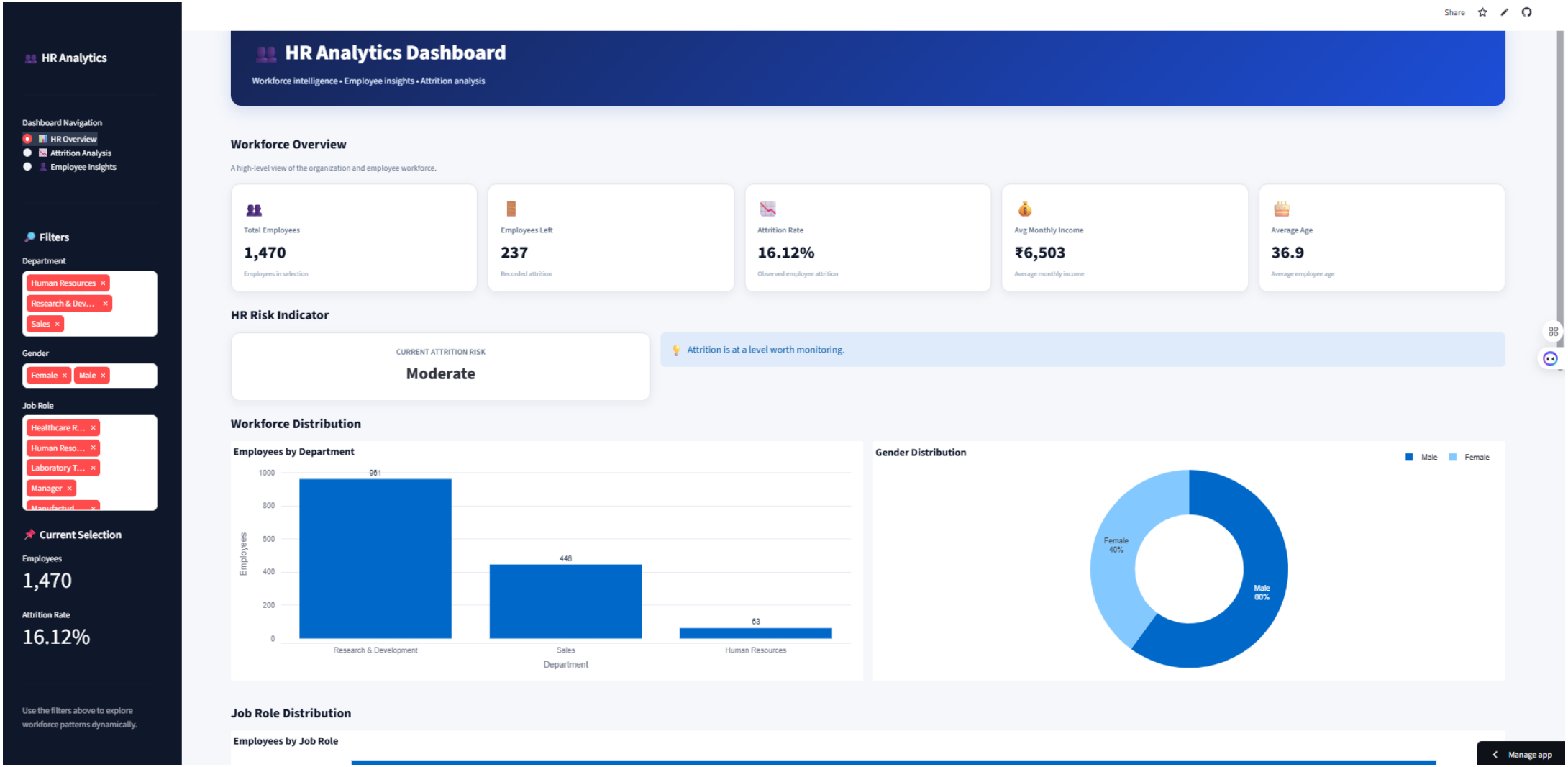
- Python — data processing and analysis
- Pandas — data manipulation and calculations
- Plotly — interactive charts
- Streamlit — dashboard development and deployment
- Git & GitHub — version control and project repository

Dashboard Sections

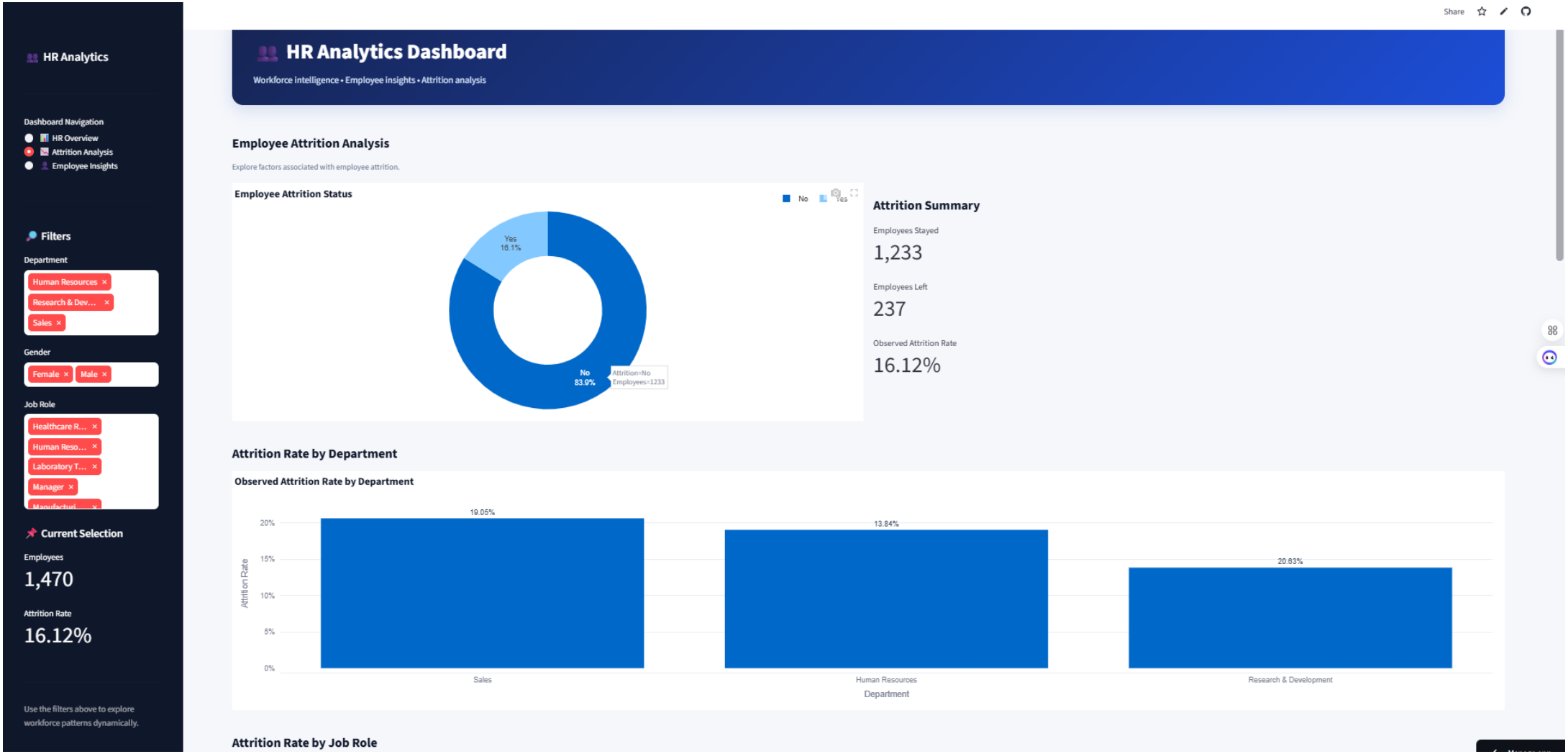
- HR Overview — workforce KPIs, department distribution, gender distribution and HR risk indicator.
- Attrition Analysis — attrition by department, job role, overtime, satisfaction, salary and tenure.
- Employee Insights — age, experience, performance, work-life balance, training and income analysis.

Filters: Department, Gender and Job Role.

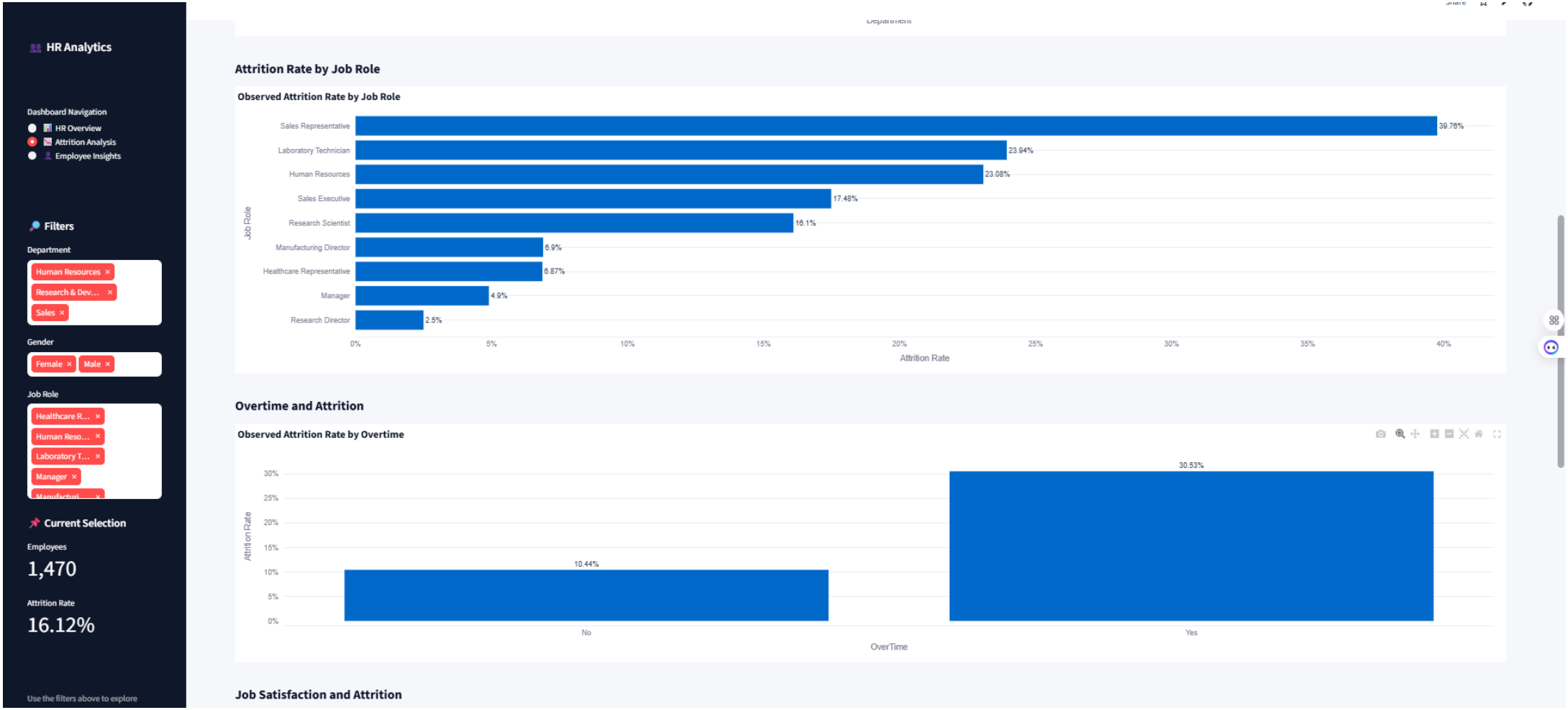
2. HR Overview — Workforce Overview



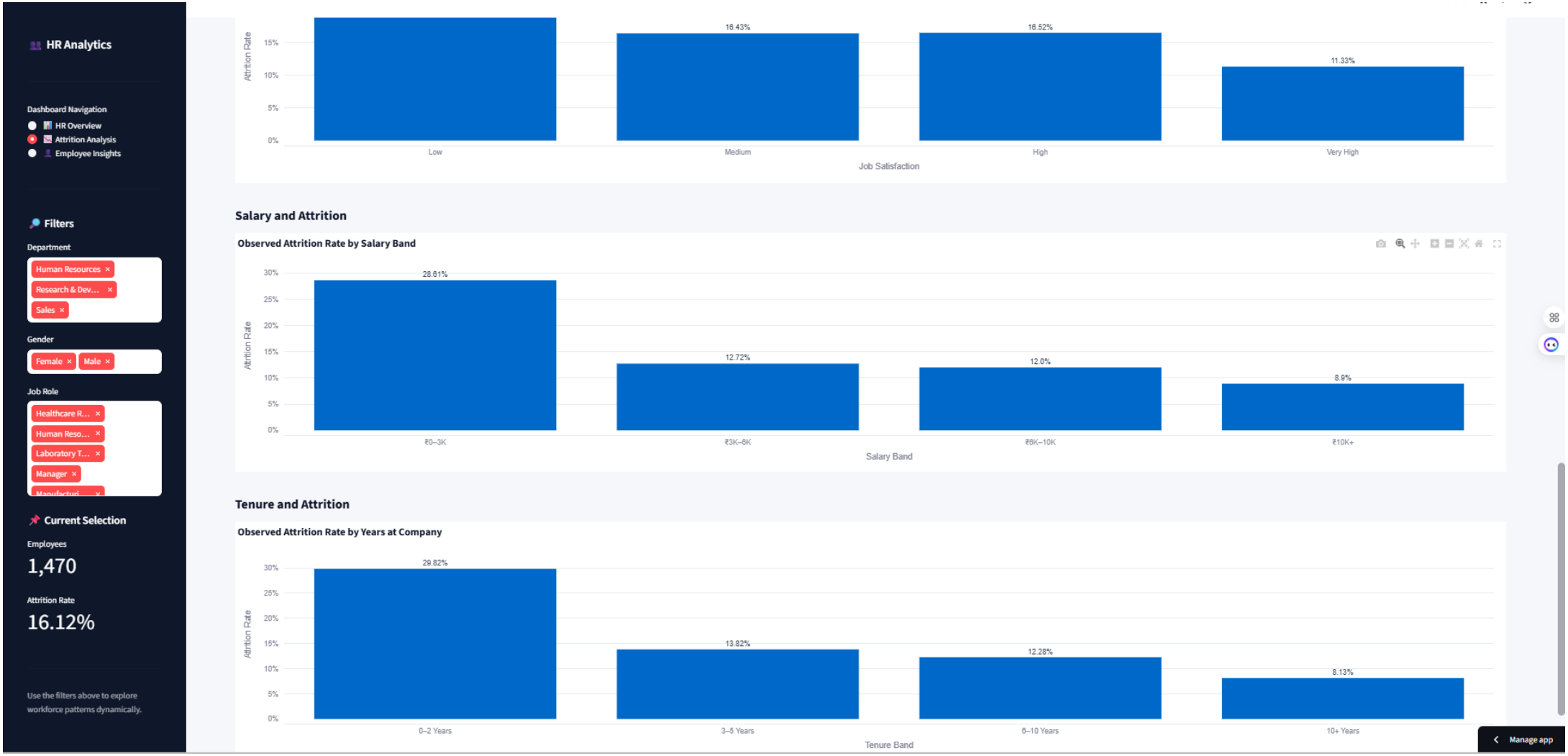
3. Attrition Analysis — Overall and Department Attrition



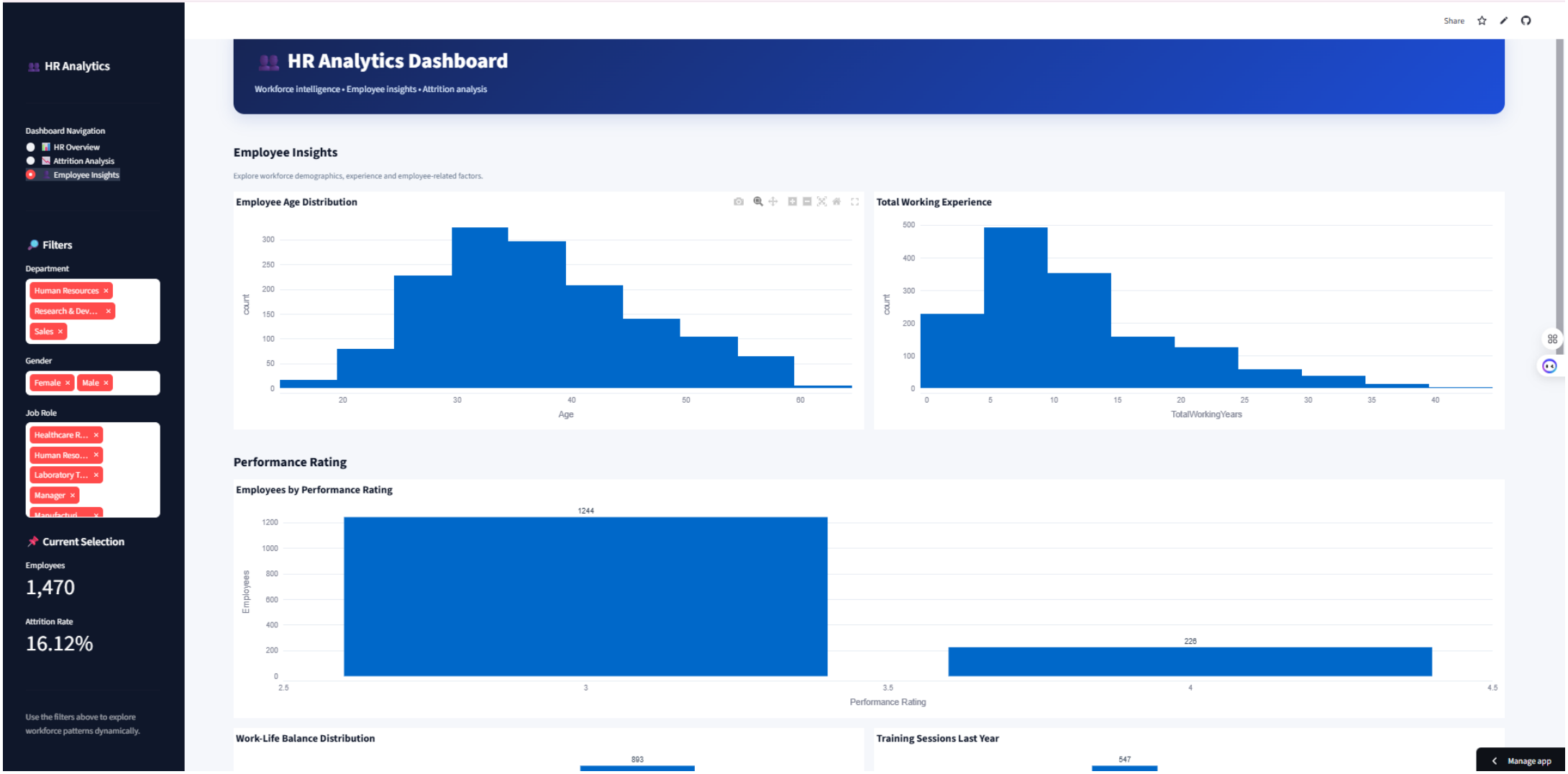
4. Attrition Analysis — Job Role and Overtime



5. Attrition Analysis — Satisfaction, Salary and Tenure



6. Employee Insights — Age, Experience and Performance



7. Employee Insights — Work-Life Balance, Training and Income



8. Key Findings

- Total employees: **1,470**; employees who left: **237**; overall attrition rate: **16.12%**.
- Sales has the highest department attrition rate at **20.63%**; Human Resources is 19.05% and Research & Development is 13.84%.
- Sales Representatives show the highest observed job-role attrition rate at approximately **39.76%**.
- Overtime employees show **30.53%** observed attrition compared with **10.44%** for employees without overtime.
- The ■0–3K salary band has the highest observed attrition at **28.61%**.
- Employees with 0–2 years at the company show the highest observed tenure-based attrition at **29.82%**.

9. Conclusion

The dashboard transforms employee-level HR data into an interactive decision-support view. It highlights workforce structure and identifies employee segments associated with higher attrition, helping HR teams focus retention and workforce-planning efforts.

Live Dashboard: harshithahranalyticsdashboard.streamlit.app

GitHub: github.com/Harshi1904/HR_Analytics_Dashboard